Q1. a) The original problem :

maximize $50x_1 + 40x_2$
subject to $2x_1 + x_2 \leq 100$
$x_1 + 3x_2 \leq 90$
$x_1, x_2 \geq 0$

∴ The dual problem :

minimize $100p_1 + 90p_2$
subject to $2p_1 + p_2 \geq 50$
$p_1 + 3p_2 \geq 40$
$p_1, p_2 \geq 0$

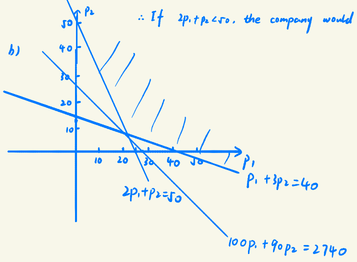
Suppose we want to buy all resources from the company

The variables : p_1 and p_2 represent the unit price of the resources.

The constraints : 2 unit of resource 1 and 1 unit of resouce 2 can produce one unit of product P_1 which is worth \$ 50
1 unit of resource 1 and 3 unit of resouce 2 can produce one unit of product P_2 which is worth \$ 40

∴ If $2p_1 + p_2 < 50$, the company would rather produce by itself.

b)



According to the graph.

The optimal value is 2740, with $\begin{bmatrix}p_1\\p_2\end{bmatrix} = \begin{bmatrix}22\\6\end{bmatrix}$

c) ∵ $b^Ty = (Ax)^Ty = x^T(A^Ty) \geq c^Tx$, as $x \geq 0$ and $A^Ty \geq c$, for $y = \begin{bmatrix}p_1\\p_2\end{bmatrix}$

To verify : For $x = \begin{bmatrix}x_1\\x_2\end{bmatrix} = \begin{bmatrix}10\\10\end{bmatrix}$, $2x_1 + x_2 = 30 \leq 100$, $x_1 + 3x_2 = 40 \leq 90$, satisfies the constraints

$50x_1 + 40x_2 = 900$

For $y = \begin{bmatrix}p_1\\p_2\end{bmatrix} = \begin{bmatrix}20\\10\end{bmatrix}$, $2p_1 + p_2 = 50 = 50$, $p_1 + 3p_2 = 50 > 40$, satisfies the constraints

$100x_1 + 90x_2 = 2900$

∴ $2900 > 900$, ∴ The Weak Duality Theorem holds.

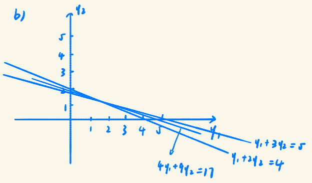
Q2. a) The primal problem :

minimize $4x_1 + 5x_2$
subject to $x_1 + x_2 \geq 4$
$2x_1 + 3x_2 \geq 9$
$x_1, x_2 \geq 0$

∴ The dual problem :

maximize $4y_1 + 9y_2$
subject to $y_1 + 2y_2 \leq 4$
$y_1 + 3y_2 \leq 5$
$y_1, y_2 \geq 0$

b)



∴ The optimal value is 17

with $y = \begin{bmatrix}y_1\\y_2\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}$

c) ∵ The optimal solution of the primal is $\begin{bmatrix}x_1^*\\x_2^*\end{bmatrix} = \begin{bmatrix}3\\1\end{bmatrix}$

∴ The objective value is $4 \times 3 + 5 \times 1 = 17$

∴ The primal and dual objective values are equal,

Q3. a) The primal problem :

minimize $5x_1 + 4x_2$
subject to $3x_1 + x_2 \geq 30$
$2x_1 + 4x_2 \geq 40$
$x_1, x_2 \geq 0$

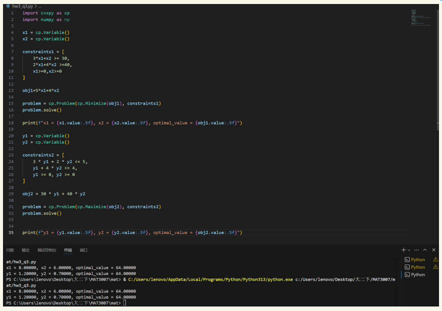
b) The dual problem :

maximize $30y_1 + 40y_2$
subject to $3y_1 + 2y_2 \leq 5$
$y_1 + 4y_2 \leq 4$
$y_1, y_2 \geq 0$

Interpretation : y_1 : the value of 1 Kg Nutrient A.
y_2 : the value of 1Kg Nutrient B

$3y_1 + 2y_2 \leq 5$, $y_1 + 4y_2 \geq 4$ ⇒ The value of Nutrient in fertilizers should be higher than the cost. of fertilizer.

(c)



```python
import cvxpy as cp
import numpy as np

x1 = cp.Variable()
x2 = cp.Variable()

constraints1 = [
    3*x1+x2 >= 30,
    2*x1+4*x2 >=40,
    x1>=0,x2>=0
]

obj1=5*x1+4*x2

problem = cp.Problem(cp.Minimize(obj1), constraints1)
problem.solve()

print(f"x1 = {x1.value:.5f}, x2 = {x2.value:.5f}, optimal_value = {obj1.value:.5f}")

y1 = cp.Variable()
y2 = cp.Variable()

constraints2 = [
    3 * y1 + 2 * y2 <= 5,
    y1 + 4 * y2 <= 4,
    y1 >= 0, y2 >= 0
]

obj2 = 30 * y1 + 40 * y2

problem = cp.Problem(cp.Maximize(obj2), constraints2)
problem.solve()


print(f"y1 = {y1.value:.5f}, y2 = {y2.value:.5f}, optimal_value = {obj2.value:.5f}")
```

```
at/hw3_q3.py
x1 = 8.00000, x2 = 6.00000, optimal_value = 64.00000
y1 = 1.20000, y2 = 0.70000, optimal_value = 64.00000
PS C:\Users\lenovo\Desktop\大二下\MAT3007\mat> & C:/Users/lenovo/AppData/Local/Programs/Python/Python313/python.exe c:/Users/lenovo/Desktop/大二下/MAT3007/m
at/hw3_q3.py
x1 = 8.00000, x2 = 6.00000, optimal_value = 64.00000
y1 = 1.20000, y2 = 0.70000, optimal_value = 64.00000
PS C:\Users\lenovo\Desktop\大二下\MAT3007\mat> 
```

∴ Both optimal solutions are 64

∴ The solutions satisfy the Strong Duality Theorem.

Q4. (a) The primal problem:

minimize $6x_1+8x_2+10x_3+7x_4+5x_5+9x_6$

subject to $x_1+x_2+x_3\leq 500$

$x_4+x_5+x_6\leq 400$

$x_1+x_4\geq 300$

$x_2+x_5\geq 400$

$x_3+x_6\geq 200$

$x_1, x_2, x_3, x_4, x_5, x_6\geq 0$

(b) The dual problem:

maximize $500y_1+400y_2+300y_3+400y_4+200y_5$

subject to $y_1+y_3\leq 6$

$y_1+y_4\leq 8$

$y_1+y_5\leq 10$

$y_2+y_3\leq 7$

$y_2+y_4\leq 5$

$y_2+y_5\leq 9$

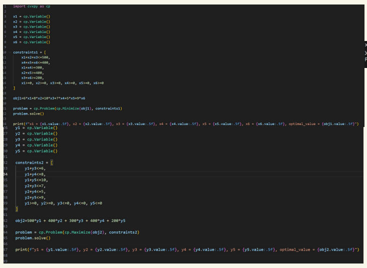
$y_1, y_2\geq 0$

$y_3, y_4, y_5\leq 0$

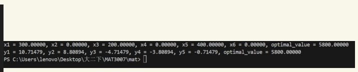
Interpretation: y_1, y_2: marginal benefit of increasing the supply capacity of W_1 and W_2

y_3, y_4, y_5: marginal cost of increasing the demand of S_1, S_2, S_3

(c)



```python
import cvxpy as cp

x1 = cp.Variable()
x2 = cp.Variable()
x3 = cp.Variable()
x4 = cp.Variable()
x5 = cp.Variable()
x6 = cp.Variable()

constraints1 = [
    x1+x2+x3<=500,
    x4+x5+x6<=400,
    x1+x4>=300,
    x2+x5>=400,
    x3+x6>=200,
    x1>=0, x2>=0, x3>=0, x4>=0, x5>=0, x6>=0
]

obj1=6*x1+8*x2+10*x3+7*x4+5*x5+9*x6

problem = cp.Problem(cp.Minimize(obj1), constraints1)
problem.solve()

print(f"x1 = {x1.value:.5f}, x2 = {x2.value:.5f}, x3 = {x3.value:.5f}, x4 = {x4.value:.5f}, x5 = {x5.value:.5f}, x6 = {x6.value:.5f}, optimal_value = {obj1.value:.5f}")
y1 = cp.Variable()
y2 = cp.Variable()
y3 = cp.Variable()
y4 = cp.Variable()
y5 = cp.Variable()

constraints2 = [
    y1+y3<=6,
    y1+y4<=8,
    y1+y5<=10,
    y2+y3<=7,
    y2+y4<=5,
    y2+y5<=9,
    y1>=0, y2>=0, y3<=0, y4<=0, y5<=0
]

obj2=500*y1 + 400*y2 + 300*y3 + 400*y4 + 200*y5

problem = cp.Problem(cp.Maximize(obj2), constraints2)
problem.solve()

print(f"y1 = {y1.value:.5f}, y2 = {y2.value:.5f}, y3 = {y3.value:.5f}, y4 = {y4.value:.5f}, y5 = {y5.value:.5f}, optimal_value = {obj2.value:.5f}")
```



```
x1 = 300.00000, x2 = 0.00000, x3 = 200.00000, x4 = 0.00000, x5 = 400.00000, x6 = 0.00000, optimal_value = 5800.00000
y1 = 10.71479, y2 = 8.80894, y3 = -4.71479, y4 = -3.80894, y5 = -0.71479, optimal_value = 5800.00000
PS C:\Users\lenovo\Desktop\大二下\MAT3007\mat> 
```

∴ The Strong Duality Theorem holds.

Q5. a) The primal problem:

minimize b^Ty

subject to $A^Ty\geq 0$

∴ The dual problem:

maximize 0^Tx

subject to $Ax=b$,

$x\geq 0$

(b) (1) According to the Weak Duality Theorem:

For feasible x, y, there must be : $0^T\leq b^Ty$

∵ If there exists x satisfing $Ax=b$ and $x\geq 0$, x is feasible

For feasible y satisfies $b^Ty<0$

$\Rightarrow b^Ty<0^Tx \Rightarrow$ It causes contradiction.

∴ If $b^Ty<0$ for a feasible y, there can't exist x such that $Ax=b$ and $x\geq 0$

(2) ∵ According to the content, $b^Ty\geq 0$

∴ b^Ty is bounded by 0, ∴ the optimal value of b^Ty is at least 0

∴ According to the Strong Duality Theorem, the optimal value of dual problem is at least 0

∴ There exists a feasible x satisfying $Ax=b$ and $x\geq 0$

(c) A: There exists $x\in\mathbb{R}^n$ such that $Ax=b$ and $x\geq 0$

B: There exists $y\in\mathbb{R}^m$ such that $A^Ty\geq 0$ and $b^Ty<0$

∴ (b) (1) can be written as $B\rightarrow$ not A

(2) can be written as not $B\rightarrow A$

∴ According to the logic, from $B\rightarrow$ not A, we can know that $A\rightarrow$ not B.

∵ $A\rightarrow$ not B, not $B\rightarrow A$

∴ $A\rightleftarrows$ not B

∴ $A\rightleftarrows$ not B, $B\rightleftarrows$ not A

∴ A, B can't be true at the same time. A, B can't be false at the same time.

∴ Exactly one of the two assertions (A, B) is true